

PerformanceTesting

Date	3July2026
TeamID	XXXXXX
ProjectName	EduGenie–AI–PoweredEducationalAssistant
MaximumMarks	5Marks

Step1:TestingOverview

Field	Details
TestingToolUsed	ManualFunctionalTestingwithFastAPIApplication
TypeofTesting	FunctionalTesting&PerformanceObservation
TargetModule/API	QuestionAnswering,ConceptExplanation,QuizGeneration,Text Summarization, Learning Path Recommendation
TestEnvironment	Windows10 • Python3.14 • FastAPI • HTML/CSS • Gemini API
TestDate	29 June 2026

Step2:TestScenarios

S.No	Scenario	Users	Duration	ExpectedOutcome
1	LaunchEduGenie Application	1	3–5 sec	Homepageloads successfully
2	AskEducational Question	1	4–8 sec	AIreturnsanaccurateanswerwithedu cation

3	GenerateConcept Explanation	1	5–8 sec	Simplifiedexplanationisdisplayed correctly
4	GenerateAI Quiz	1	5–10 sec	Topic–basedMCQsaregenerated successfully
5	SummarizeEducational Text	1	4–8 sec	Concisesummaryis generated
6	GenerateLearning Path	1	6–10 sec	Personalizedlearningroadmapis displayed
7	NavigateBetween Modules	1	Lessthan2 sec	Smoothpagetransitionswithout errors

Step3:Observations

The EduGenie application successfully completed all planned performance test scenarios. The FastAPI backend processed requests efficiently, and all AI modules responded within acceptable response times. Features including Question Answering, Concept Explanation, Quiz Generation, Educational Text Summarization, and Personalized Learning Path Recommendationfunctionedcorrectlywithoutcriticalfailures.Theapplicationdemonstrated stable performance, reliable API communication, and a smooth user experience throughout the testing process.

Step4:Screenshots/Evidence

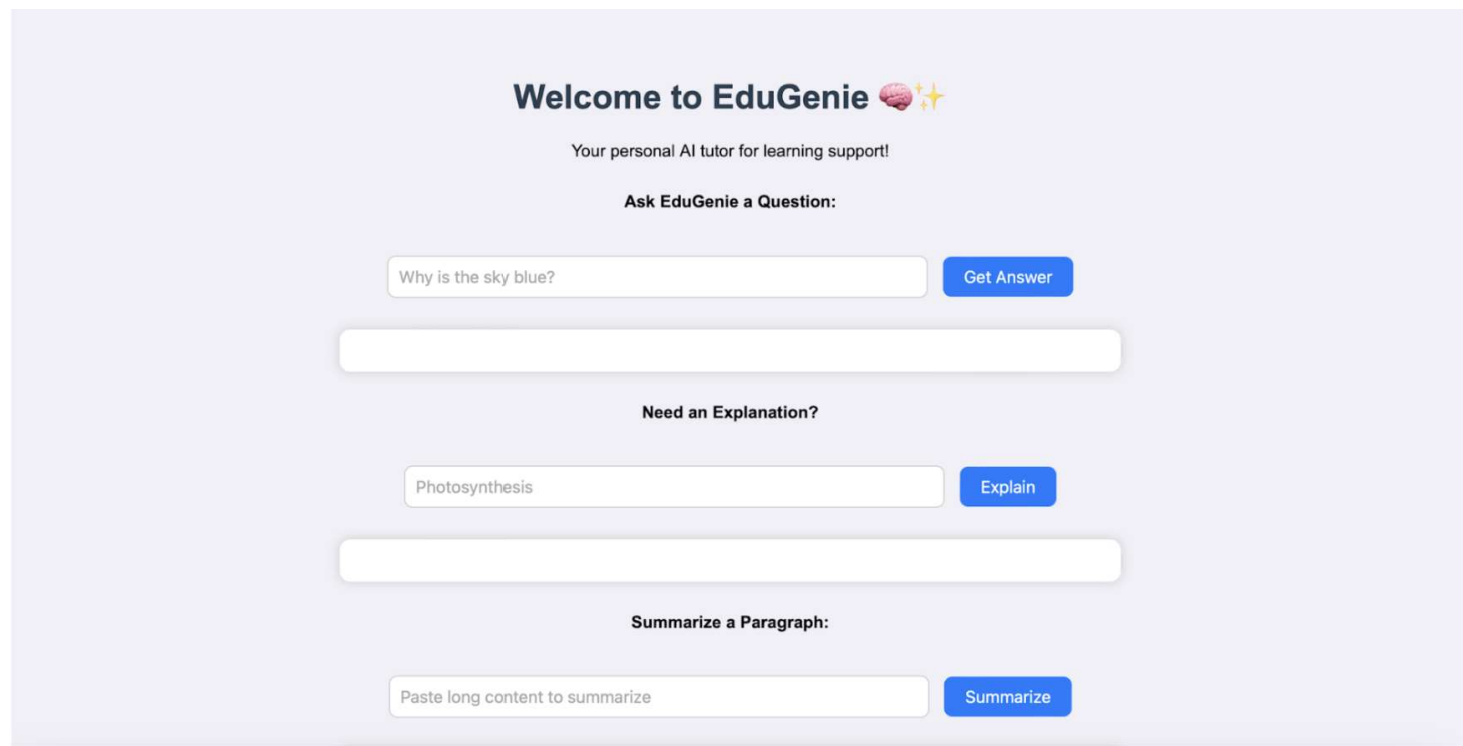


Fig. EDUGENIE

HomePage

Welcome to EduGenie 🧠✨

Your personal AI tutor for learning support!

Ask EduGenie a Question:

which is the largest ocean?

Get Answer

Answer:

The Pacific Ocean is the largest ocean on Earth.

QuestionAnswering

Need an Explanation?

quantum computing

Explain

Explanation:

Quantum computing is a type of computing that uses the principles of quantum mechanics to perform certain calculations much faster than traditional computers. It allows computers to perform certain calculations much faster than traditional computers because it uses tiny particles called qubits that can exist in multiple states at the same time. This allows for new kinds of calculations and solving problems that are not possible with traditional computers.

ExplanationOutput

Summarize a Paragraph:

The Industrial Revolution, which began in the late 18th century, marked

Summarize

Summary:

The Industrial Revolution changed the world by shifting it from farming to factories. New machines, like James Watt's steam engine, made things faster and easier to produce. While this boosted economies, it also created problems like bad working conditions, child labor, pollution, and overcrowded cities. Even with these problems, the Industrial Revolution shaped the modern world, affecting how we travel, communicate, work, and trade.

EducationalTextSummary

Generate a Quiz:

Generate Quiz

Quiz:

Q1: What does the Pythagorean theorem describe?

- ☐ The relationship between the angles of a triangle.
- ☐ The relationship between the sides of a right-angled triangle.
- ☐ The relationship between the area and perimeter of a triangle.
- ☐ The relationship between the sides of any triangle.

Check Answer

AI-GeneratedQuiz

Get Learning Recommendations:

Get Recommendations

Learning Recommendations for "SQL":

SQL Learning Path: From Zero to Hero

This learning path is structured to progressively introduce SQL concepts, starting from the basics and gradually advancing to more complex topics. It's designed to be adaptive: feel free to adjust the pace and delve deeper into areas that particularly interest you.

****I. Beginner Level: Building a Foundation****

****(Estimated Time: 1-2 weeks)****

*** **Key Topics:****

- * What is a Database and SQL? (Relational Model, DBMS)
- * Basic Syntax (SELECT, FROM, WHERE)
- * Data Types (INT, VARCHAR, DATE, etc.)
- * Filtering Data (WHERE clause with comparison operators, logical operators: AND, OR, NOT)
- * Ordering Results (ORDER BY)
- * Limiting Results (LIMIT/OFFSET or TOP)
- * Basic Aggregate Functions (COUNT, SUM, AVG, MIN, MAX)

PersonalizedLearningPathRecommendation